4. Let $V(x, y, z) = xy + yz + zx, x, y, z \in \mathbb{R}$. Find the differential $dV$.

## 8.6.1 Function of Function Rule

Let $F$ be a function of two variables $x, y$. Sometimes these variables may be functions of a single variable having same domain. In this case, the function $F$ ultimately depends only on one variable. So we should be able to treat this $F$ as a function of single variable and study about $\frac{dF}{dt}$. In fact, this is not a coincidence, it can be proved that

**Theorem 8.2**

Suppose that $W(x, y)$ is a function of two variables $x, y$ having partial derivatives $\frac{\partial W}{\partial x}, \frac{\partial W}{\partial y}$. If both the variables $x, y$ are differentiable functions of a single variable $t$, then $W$ is a differentiable function of $t$ and

$$\frac{dW}{dt} = \frac{\partial W}{\partial x}\frac{dx}{dt} + \frac{\partial W}{\partial y}\frac{dy}{dt} \qquad ...(16)$$

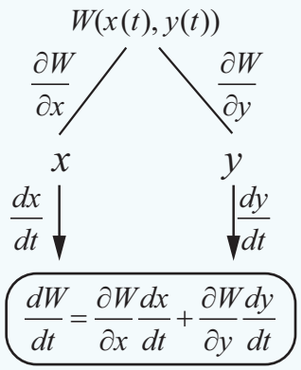



Fig. 8.14

Let us consider an example illustrating the above theorem.

### Example 8.18

Verify the above theorem for $F(x, y) = x^2 - 2y^2 + 2xy$ and $x(t) = \cos t, y(t) = \sin t, t \in [0, 2\pi]$.

### Solution

Let $F(x,y) = x^2 - 2y^2 + 2xy$ and $x(t) = \cos t$, $y(t) = \sin t$.

Then $F(x, y) = \cos^2 t - 2\sin^2 t + 2\cos t \sin t$ and thus $F$ has becomes a function of one variable $t$. So by using chain rule, we see that

$$\begin{aligned}\frac{dF}{dt} &= 2\cos t(-\sin t) - 4\sin t\cos t + 2(-\sin^2 t + \cos^2 t) \\ &= -6\cos t \sin t + 2(-\sin^2 t + \cos^2 t).\end{aligned}$$

On the other hand if we calculate

$$\begin{aligned}\frac{\partial F}{\partial x}\frac{dx}{dt} + \frac{\partial F}{\partial y}\frac{dy}{dt} &= (2x + 2y)\frac{dx}{dt} + (2x - 4y)\frac{dy}{dt} \\ &= 2(\cos t + \sin t)(-\sin t) + 2(\cos t - 2\sin t)(\cos t) \\ &= -6\cos t \sin t + 2(-\sin^2 t + \cos^2 t) \\ &= \frac{dF}{dt}.\end{aligned}$$



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